Road Accident Analysis

New York State · 2016–2023

Linear and logistic regression modelling on the US-Accidents dataset. Analysis of environmental, infrastructural, and temporal predictors of visibility and day/night accident classification.

Data Mining — A.A. 2024/2025
Dataset: US-Accidents · Sobhan Moosavi et al. · Kaggle

1. Dataset

The source dataset is **US-Accidents** (Moosavi et al.), publicly available on Kaggle, covering traffic accidents across 49 US states from February 2016 to March 2023. The analysis focuses exclusively on **New York State (NY)**. After preprocessing, the working sample was reduced from 7,730,000 observations and 43 variables to 8,448 observations and 15 variables.

Parameter	Original dataset	After preprocessing
Observations	7,730,000	8,448
Variables	43	15
States / scope	49 US states	New York (NY)
City levels	477	125
Weather_Condition levels	47	11

1.1 Variables

Variable	Type	Description
Severity	Continuous	Accident severity score (1–4)
Distance	Continuous	Road extent affected (miles)
City	Categorical	City of accident — 125 levels after grouping
Temperature	Continuous	Temperature in °F
Pressure	Continuous	Atmospheric pressure (inches Hg)
Visibility *	Continuous	Target variable — visibility in miles
Wind_speed	Continuous	Wind speed (mph)
Weather_Condition	Categorical	Weather at time of accident — 11 levels after grouping
Crossing	Binary	Near a pedestrian crossing (True/False)
Junction	Binary	Near an intersection (True/False)
Stop	Binary	Near a stop sign (True/False)
Traffic_signal	Binary	Near a traffic light (True/False)
Sunrise_Sunset	Categorical	Day / Night — also binary target for logistic model
Month	Categorical	Month of accident (1–12)
Day	Categorical	Day of month (1–31)

** Primary response variable for the linear regression model.*

1.2 Categorical variable preprocessing

City: Cities with fewer than 20 recorded accidents were grouped into Other and subsequently removed, reducing levels from 477 to 125. **Weather_Condition:** Optimal grouping via the *factorMerger* package (GIC criterion) reduced levels from 47 to 11, balancing granularity and parsimony.

2. Missing Value Analysis

Missing values were identified using the *VIM* package (aggr function). Single imputation was applied: **mean imputation** for continuous variables and **mode imputation** for categorical ones. Correct imputation was verified graphically with a second aggr plot post-imputation.

Variable	Type	Imputation method
wind_speed	Continuous	Mean — Hmisc::impute
temperature	Continuous	Mean — Hmisc::impute
pressure	Continuous	Mean — Hmisc::impute
visibility	Continuous	Mean — Hmisc::impute
Weather_Condition	Categorical	Mode — Hmisc::impute
Sunrise_Sunset	Categorical	Mode — Hmisc::impute

3. Linear Model — Target: Visibility

The response variable is **visibility** (miles). Visibility was chosen as the target because it directly captures environmental accident risk: understanding how atmospheric and infrastructural predictors affect visibility can inform road-safety policy and signage improvements. A shift of +1 was applied to all visibility values to ensure strict positivity before Box-Cox testing.

3.1 Model 1 — Full baseline

All 15 variables were included. Non-significant predictors: *Severity, Crossing, Stop, Traffic_signal, Temperature, Sunrise_Sunset*. $R^2 = 0.757$ | AIC = 33,159.

3.2 Multicollinearity check

Quantitative variables: VIF < 5 and TOL > 0.04 for all predictors — no collinearity detected. Pair panels (psych) confirmed the result graphically. **Qualitative variables:** Normalised chi-square < 0.80 for all pairs — no collinearity among factors.

3.3 Interactions — Model 2

The interaction **temperature × Sunrise_Sunset** is statistically significant. Adding it improves AIC from 33,159 to 33,153 and R^2 from 0.7573 to 0.7575.

3.4 Response transformation — Box-Cox

Box-Cox suggested $\lambda = 1.67$. After transforming the response, AIC deteriorated sharply (33,153 → 64,090) and R^2 decreased. The response was therefore **not transformed**.

3.5 Predictor transformations — GAM (Model 4)

Loess and Spline fits via the *gam* package indicated a **cubic relationship** for both *distance* and *pressure*. Cubic polynomial terms were added. The RESET test confirmed their relevance (p-value < 2.2e-16). R^2 improved to 0.760.

3.6 Model selection — stepAIC (Model 5)

Bidirectional stepAIC was applied to Model 4. The selected model retains only significant predictors. AIC improved to 33,079; $R^2 = 0.760$.

3.7 Outlier removal — DFFITS (Model 6)

Influential observations were identified using the DFFITS criterion: $|DFFITS| > 2\sqrt{p/n}$. Their removal produced a substantial improvement: R^2 rose from 0.760 to **0.851**, AIC dropped to **26,673**.

3.8 Heteroscedasticity

Heteroscedasticity was detected via *ncvTest* (car) and the Breusch-Pagan test (p-value = 2.22e-16). Robust inference was obtained using **White's robust standard errors** (sandwich package, vcovHC).

3.9 Bootstrap validation — R = 1,999 replications

Bootstrap resampling confirmed the robustness of all estimated parameters. Confidence intervals (95%, percentile method) are narrow and symmetric for the main effects, with slight instability only in some *Weather_group* levels that are also non-significant in the model summary. The final model is stable.

3.10 Model performance summary

Model	R^2	AIC	Notes
Model 1 — Full baseline	0.757	33,159	All 15 predictors
Model 2 — + Interaction	0.757	33,153	temperature × Sunrise_Sunset
Model 4 — + Cubic transforms	0.760	33,083	$distance^3$, $pressure^3$
Model 5 — stepAIC	0.760	33,079	Parsimonious selection
Model 6 — Outlier removal (final)	0.851	26,673	DFFITS + robust SE

3.11 Final model specification

```
modello_final <- lm(
visibility ~ distance + I(distance^2) + I(distance^3)
+ City
+ pressure + I(pressure^2) + I(pressure^3)
+ wind_speed + Weather_group + Severity
+ month + day
+ temperature * Sunrise_Sunset,
data = b_filtered
)

# Robust inference
coeftest(modello_final, vcov = vcovHC(modello_final))
```

$R^2 = 0.851$ | $R^2_{adj} = 0.836$ | AIC = 26,673

4. Logistic Model — Target: Sunrise/Sunset (Day vs. Night)

A logistic regression model (GLM, family = binomial) was built to classify whether an accident occurred during the day or at night (binary variable *Sunrise_Sunset*).

4.1 Modelling pipeline

Full logistic model: GLM on all available predictors. Non-significant variables: *Junction, Traffic_signal, visibility*.

stepAIC selection: The selected model (AIC = 10,127.52) retains: *Severity, distance, City, Crossing, Junction, Stop, month, day, wind_speed, temperature, pressure, visibility, Weather_group*.

ANOVA / LRT comparison: The likelihood-ratio test between the full and the selected model yields p = 0.5994 (> 0.05), deviance change = –0.33. The parsimonious model performs equivalently to the full one.

4.2 Odds ratio interpretation

The odds ratio (OR) measures the association between each predictor and the probability that an accident occurred during the **day**. OR = 1: no association; OR > 1: associated with day accidents; OR < 1: associated with night accidents.

Predictor / Group	OR	Interpretation
Amsterdam, Huntington Station, Fresh Meadows	> 1	Strongly increases probability of Day
Weather group 6	> 1	Increases probability of Day accidents
Hudson, Kirkville, Jordan	< 1	Associated with Night accidents
Reduced visibility	< 1	Associated with Night accidents
Wind speed, Temperature	≈ 1	No significant effect on day/night classification

5. R Packages

Package	Purpose
dplyr	Data manipulation and filtering
VIM	Missing value visualisation (aggr)
Hmisc	Missing value imputation (mean / mode)
factorMerger	Optimal grouping of categorical levels (GIC criterion)
mctest	VIF / TOL for multicollinearity diagnostics
psych	Pair panels for exploratory correlation analysis
gam	Generalised additive models — Loess and Splines
lmtest	RESET test and Breusch-Pagan heteroscedasticity test
MASS	stepAIC model selection and Box-Cox transformation
car	ncvTest, bootstrap (Boot)
sandwich	White's robust standard errors (vcovHC)

6. Replication Instructions

To reproduce the full analysis from scratch:

1. Download the **US-Accidents** dataset from Kaggle (Sobhan Moosavi et al., 2023).
2. Filter the dataset for State == 'NY' to obtain the New York subset.
3. Set the working directory in DM_code.R via setwd().
4. Install all packages listed in Section 5.
5. Run DM_code.R sequentially — each block is commented to guide the analysis.

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